

MOSES OMOTO

ALL-IN-ONE MASTER PROFESSIONAL CV

Cybersecurity Specialist | Software Engineer | IT Systems & Helpdesk Administrator | ICT & Data Officer

Nairobi, Kenya | Phone: +254 791 688 623 / +254 112 645 206 | Email: mosesomoto@gmail.com

LinkedIn: linkedin.com/in/moses-omoto-adala | GitHub: github.com/moses-maker | Portfolio: moses-maker.github.io

1. EXECUTIVE PROFESSIONAL SUMMARY

Versatile, results-driven Technology Specialist and Departmental Leader with a Bachelor of Science in Computer Security & Forensics and over 3 years of combined experience spanning Cybersecurity Operations (DevSecOps, Penetration Testing, Threat Analysis), Full-Stack Software Engineering (Python/Django, Node.js, REST APIs), IT Infrastructure & Helpdesk Support (50+ Workstations), and Digital Data Management. Currently serving as Lecturer and Head of Department for ICT & Engineering, managing departmental operations, network environments, laboratory readiness, and student project supervision. Proven track record of bridging offensive security methodologies with defensive software development, automating complex data pipelines, standardizing preventive IT maintenance (reducing repair incidents by ~50%), and training 200+ technical learners annually with a 94% practical competency rate. Fully equipped for high-impact international remote, hybrid, or on-site roles across technology disciplines.

2. MASTER TECHNICAL & CORE COMPETENCIES

Cybersecurity & DevSecOps	Penetration Testing, Vulnerability Assessment & Remediation, DevSecOps & SDLC Integration, OWASP Top 10 Mitigation, Static/Dynamic Code Review, Threat Analysis & Event Logging, OSINT, Privilege Escalation, Applied Cryptography Concepts, Security Auditing.
Software & Backend Engineering	Python (Django, Flask, Automation), JavaScript (Node.js, Express.js), RESTful API Architecture, HTML5/CSS3, Microservices, MVC Architecture, Test-Driven Development (TDD), Docker Containerization, CI/CD Security Automation (Jenkins, GitHub Actions).
IT Support & Systems Administration	1st/2nd-Line Helpdesk Support, Workstation Provisioning & Deployment (50+ PCs), Windows 10/11 & Linux (Ubuntu, Kali) Administration, Endpoint Security & Patch Management, Network Troubleshooting, Peripheral/Hardware Diagnostics, Jira & Trello Support Ticketing.
Data Operations & Digital Health	Data Collection Tools, KHIS (DHIS2) & Digital Health Platforms, Python Data Extraction (Selenium, BeautifulSoup), Interactive Analytics Dashboards (Streamlit, Pandas, NumPy), SQL & NoSQL Databases (PostgreSQL, MySQL, MongoDB, Redis), M&E Data Review Preparation.
AI, Machine Learning & Education	Natural Language Processing (NLP), Scikit-Learn, TensorFlow, Prompt Engineering, AI Model Evaluation & Dataset Annotation, Curriculum Development, Student Mentorship, Technical Documentation, Asynchronous Team Leadership.

3. TECHNICAL ENVIRONMENT & TOOLS TOOLKIT

- **Languages & Web:** Python, JavaScript (ES6+), HTML5, CSS3, SQL, Bash Scripting
- **Frameworks & Libraries:** Django, Node.js, Express.js, Streamlit, Flask, Pandas, NumPy, Scikit-Learn, TensorFlow, Bootstrap
- **Security & Pentesting Tools:** Nmap, Burp Suite, Wireshark, Metasploit, Gobuster, SQLmap, Hydras, VulnHub Labs, TryHackMe
- **DevOps, Cloud & DBs:** AWS (EC2/S3), Docker, Git, GitHub, GitHub Actions, Jenkins, PostgreSQL, MySQL, MongoDB, Redis
- **Operating Systems & Virtualization:** Linux (Ubuntu, Kali, Linux Mint), Windows 10/11 & Server, VMware, Hyper-V, VirtualBox
- **Productivity & Collaboration:** Jira, Trello, Asana, Notion, Slack, Zoom, Google Workspace, Microsoft Teams, VS Code

4. PROFESSIONAL EXPERIENCE

LECTURER & HEAD OF DEPARTMENT – ICT & ENGINEERING

ICS Technical College, Kenya | 2024 – Present (Remote-Hybrid)

- Manage, configure, and maintain 50+ Windows and Linux workstations and networking hardware across computing laboratories, ensuring 99%+ operational availability for 200+ students, lecturers, and staff.
- Deliver first- and second-line IT helpdesk support using Jira/Trello ticketing workflows, diagnosing and resolving hardware, operating system, network connectivity, and application issues promptly.
- Lead departmental technical operations and practical instruction in Cybersecurity, Software Engineering (Python/JavaScript), Linux Systems, Operating Systems, and Networking.
- Engineered standardized preventive maintenance practices and software update workflows across departmental laboratories, achieving an approximate 50% reduction in equipment repair incidents.
- Architected and deployed a Django-based threat analysis web application used as a live teaching and assessment platform, implementing role-based access controls and OWASP Top 10 remediations (XSS, CSRF, input validation).
- Supervise and evaluate 30+ student engineering and cybersecurity projects annually, assessing system architecture, code quality, security vulnerability controls, and documentation.
- Instruct students on secure software development lifecycle (SDLC) practices, static/dynamic code reviews, automated testing, and ethical penetration testing lab exercises (reconnaissance, enumeration, privilege escalation).
- Introduced project-based learning methodologies, improving student course engagement by 40% and project completion rates by 35% (achieving a 94% practical competency rate).

LECTURER / HOD ICT & TECHNICAL TRAINING COORDINATOR

East Africa Institute of Certified Studies, Kenya | Sep 2022 – Present

- Coordinated technical operations, computer laboratory readiness, and practical training across ICT, computer science, network administration, and database management programs.
- Supported ICT laboratory maintenance and technical troubleshooting initiatives that contributed to a ~50% reduction in hardware repair incidents and improved machine uptime.
- Supervised student projects, evaluated technical competencies in web development and systems administration, and provided constructive feedback to ensure project success.
- Created structured technical learning materials, practical assessment guides, incident log documentation, and quality control reports.

FREELANCE FULL-STACK DEVELOPER & CYBERSECURITY / DATA CONSULTANT

Self-Employed | 2022 – 2024 (Remote)

- Engineered and deployed 5+ secure full-stack web applications using Django, Node.js, Express, and PostgreSQL/MongoDB for regional and international clients, managing projects end-to-end remotely.
- Conducted penetration testing and vulnerability assessments on client web platforms, identifying and remediating OWASP Top 10 security flaws (SQLi, XSS, CSRF, authentication bypass).

- Programmed automated web data extraction pipelines using Python, Selenium, and Pandas, reducing manual data processing time by 85% while enforcing data validation rules.
- Architected interactive Streamlit analytics dashboards for data visualization and product reporting, containerized via Docker and deployed on AWS EC2.
- Managed remote CI/CD automation pipelines using GitHub Actions and Jenkins, cutting deployment cycle times by 60% across distributed client environments.

5. FEATURED TECHNICAL PROJECTS & SECURITY LABS

- **Threat Analysis Web Application (Django) | Full-Stack Security Engineering:** Designed and developed a production-ready security threat analysis platform using Django, PostgreSQL, and Bootstrap. Built modules for security event logging, threat detection, and risk reporting with role-based access control (RBAC). Implemented protections against XSS, CSRF, and input validation vulnerabilities. Deployed on AWS with Docker containerization.
- **End-to-End Penetration Testing & Ethical Hacking Labs (VulnHub / CTFs):** Executed full penetration testing assessments across simulated enterprise Linux environments (DC-7, Nightfall, HackInOS, Kioptrix). Conducted reconnaissance (Nmap, OSINT), web application directory enumeration (Gobuster, Burp Suite), vulnerability discovery, exploitation (RCE, reverse shells), and privilege escalation via misconfigured cron jobs and SUID binaries to achieve root access. Produced published video walkthroughs and technical remediation reports.
- **Automated Data Extraction & Scraper Pipeline | Python, Selenium, Pandas:** Programmed an automated data extraction system utilizing Python and Selenium to scrape, clean, and structure dynamic web datasets (IMDb, dynamic portals). Reduced manual collection time by 85% while exporting structured CSV/JSON payloads for reporting pipelines.
- **Interactive Data Analytics & Streamlit Dashboards | Python, Streamlit, SQL:** Built responsive, web-based analytics dashboards for real-time dataset visualization, product metrics presentation, and NLP sentiment analysis (achieving 87% accuracy using Scikit-Learn TF-IDF vectorization).

6. EDUCATION & ACADEMIC BACKGROUND

BACHELOR OF SCIENCE IN COMPUTER SECURITY & FORENSICS (2022)

Jaramogi Oginga Odinga University of Science and Technology, Kenya

- Final Year Project: Threat Analysis Web Application (Django)
- Key Coursework: Computer Security, Digital Forensics, Secure Systems Development, Network Security, Database Systems, Operating Systems, Ethical Hacking.

7. CERTIFICATIONS & TECHNICAL TRAINING

- Python Developer Certification — SoloLearn (Certificate ID: CC-DNGAQW3V)
- Ethical Hacking & Penetration Testing — Practical Security Labs (VulnHub: DC-7, Nightfall, HackInOS, Kioptrix; TryHackMe; Hack The Box)
- Full-Stack Web Development — Python, JavaScript, HTML5, CSS3, Node.js & Express
- Google Workspace Administration — IT System Administration & Security Policies (LinkedIn Learning)
- AWS Cloud Practitioner Principles — Cloud Infrastructure & Deployment (In Progress)

8. ADDITIONAL INFORMATION & GLOBAL AVAILABILITY

- **Work Environment & Availability:** Fully setup for immediate full-time remote, hybrid, or on-site roles worldwide. Equipped with high-speed fiber internet and dedicated home office infrastructure.
- **Timezone Flexibility:** Based in Nairobi (EAT, UTC+3). Full timezone overlap for European (CET/GMT) teams and EST/EDT evening collaboration schedules.
- **Languages:** Fluent in English (Native/Bilingual professional proficiency) and Swahili (Native). Strong technical documentation and presentation skills.

- **Field Mobility & Safeguarding:** Willing and available for extensive field travel and remote deployments. Knowledgeable in inclusive programming, disability inclusion, and data safeguarding principles.